3 **(a)** work done is the product of force and the distance <u>moved</u> in the direction of the force
or product of force and displacement in the direction of the force B1 [1]

(b)	**(i)**		work done equals the <u>decrease</u> in GPE – <u>gain</u> in KE	B1	[1]
	(ii)	**1.**	distance = area under line	C1	
			= (7.4 × 2.5) / 2 = 9.3 m (9.25 m)	M1	[2]
			or		
			acceleration from graph a = 7.4 / 2.5 (= 2.96)	(C1)	
			and equation of motion $(7.4)^2 = 2 \times 2.96 \times s$ gives s = 9.3 (9.25) m	(A1)	
		2.	kinetic energy = $\frac{1}{2} m v^2$	C1	
			$= \frac{1}{2} \times 75 \times (7.4)^2$	C1	
			= 2100 J	A1	[3]
		3.	potential energy = mgh	C1	
			h = 9.3 sin 30°	C1	
			PE = 75 × 9.81 × 9.3 sin 30° = 3400 J	A1	[3]
		4.	work done = energy loss	C1	
			R = (3421 – 2054) / 9.3	C1	
			= 150 (147) N	A1	[3]